MILITARY SPECIFICATION

H M X FOR 30MM AMMUNITION

1. SCOPE

1.1 This specification covers two types of explosive designated as Explosive, HMX, PL Number I and Explosive, HMX, PL Number II. (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-G-155	-	Graphite
JAN-C-263	-	Calcium Stearate
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-C-20470	-	Calcium Resinate
MIL-H-45444	-	H M X

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage

FSC: 1376

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MIL-STD-650	-	Explosive: Sampling, Inspection and Testing
MIL-STD-1171	-	Acceptance and Description Sheets for Propellant and Explosive
MIL-STD-1168	-	Lot Numbering of Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0 - 190

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-The explosive, HMX, PL Number I and Number II shall conform to the requirements of Table I, when determined as specified in 4.3.1-4.3.4.

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TABLE I

Explosive, HMX, PL Number I

Ingredient	Percent Composition
HMX, MIL-H-45444, Grade II, Class C	97.5 ± 0.2
Calcium Resinate, MIL-C-20470, Type II	0.5 ± 0.1
Graphite, MIL-G-155, Grade I or II	2.0 ± 0.2

Explosive, HMX, PL Number II

Ingredient	Percent Composition
HMX, MIL-H-45444, Grade II, Class A	98.0 ± 0.2
Calcium Stearate, JAN-C-263	1.5 ± 0.1
Graphite, MIL-G-155, Grade I or II	0.5 ± 0.1

3.2 Moisture.-The moisture content of the explosive composition shall not exceed 0.10 percent when determined as specified in 4.3.5.

3.3 Foreign matter.-The explosive composition shall be free of foreign matter when determined as specified in 4.3.6.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of material deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents and that certificates of analysis are on file and available for review.

d. Quantity in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of one or more batches of explosive of the same type produced from one individual lot of HMX under one continuous set of operating conditions. Each batch shall consist of that quantity of explosive which has been subjected to the same chemical or physical mixing process intended to make the final product homogeneous. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Acceptance and Description Sheets conforming to MIL-STD-1171 shall be prepared.

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4.2.2 Examinations.-The examinations listed in the following subparagraphs shall be performed on inspection lots as defined in MIL-STD-105. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified and it shall be documented as part of the contractor's inspection system.

4.2.2.1 Bag, cotton duck (See 5.1)

Categories	Defect	Method of Inspection	Defect Code No. (See 6.2)
Critical: None defined			
Major:	AQL 0.40 Percent		
101.	Foreign matter	Visual	01001
102.	Bag pierced or torn	Visual	01002
Minor: None defined			

4.2.2.2 Sealed rubber bag or suitable water-tight material (See 5.1)

Category	Defect	Method of Inspection	Defect Code
Critical: None defined			
Major:	AQL 0.40 Percent		
101.	Bag improperly closed	Visual-Manual	02001
102.	Bag damaged	Visual	02002
Minor: None defined			

4.2.2.3 Sealed keg or drum (See 5.2)

Category	Defect	Method of Inspection	Defect Code
Critical: None defined			
Major:	AQL 0.40 Percent		
101.	Sealing improper	Visual-Manual	03001
102.	Keg or drum damaged	Visual	03002
Minor:	AQL 0.65 Percent		
201.	Marking missing, improper or illegible	Visual	03003

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4.2.3 Testing

4.2.3.1 Sampling.-A random sample of eight (8) containers shall be selected from each lot. When lots are packed in eight (8) containers or less, each container shall be sampled.

4.2.3.2 Preparation of composite.-A composite sample of eight ounces shall be made by taking approximately equal quantities from each sample container. After mixing to form a homogeneous sample, it shall be subjected to the tests specified in 4.3. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.2.4 Inspection equipment.-For the performance of all tests and examinations specified in 4.2 and 4.3, commercial inspection equipment should be employed (see 6.5). The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of insuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.3 Test methods and procedures

4.3.1 Determination of calcium resinate or calcium stearate as applicable, Major defect, Defect Code No. 04001.-Transfer an accurately weighed portion of approximately 5 grams of the composite to a tared medium porosity sintered glass crucible. Extract the sample with three 20 millilites (ml) portions of chloroform which has been previously saturated with HMX, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Stir occasionally to break up any lumps which may persist. Wash down the sides of the crucible with each 20 ml portion of chloroform to insure complete removal of the calcium resinate or stearate. Aspirate the residue until the odor of chloroform is no longer detectable. Place the crucible and contents in a drying oven at 100° to 105° Centigrade (100°-105°C) for one half hour. Cool in a desiccator and weigh. Repeat the above extraction, drying and weighing until the loss in weight on subsequent extraction totals less than 0.003 gm. Calculate the percent of calcium resinate or stearate on a moisture-free basis as follows:

$$\text{Percent Calcium Resinate or Stearate} = \frac{100A}{W}$$

Where

A = Loss in weight of the crucible and contents
W = Weight of specimen on a moisture-free basis

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4.3.2 Determination of graphite content, Major defect, Defect Code No. 05001.-Extract the residue remaining after the calcium resinate or stearate has been removed with two 20 ml portions of dimethyl formamide, maintaining each with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring, before applying suction and until the following qualitative test for HMX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml beaker, evaporate to dryness on a steam bath and add several drops of a solution of Chromotropic acid in normal (N) sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of HMX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100° to 105°C for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue to percentage of graphite on a moisture-free basis as follows:

$$\text{Percent graphite} = \frac{100A}{W}$$

Where

A = Weight of residue in tared crucible.
W = Weight of specimen on a moisture-free basis

4.3.3 Determination of total binder-lubricant content, Major defect, Defect Code No. 06001.-The percentage total binder-lubricant shall be determined by adding the percent calcium resinate or stearate to the percent graphite.

4.3.4 Determination of HMX content, Major defect, Defect Code No. 07001.-The percent HMX content shall be determined by subtracting the percent total binder-lubricant from 100 percent.

4.3.5 Determination of moisture, Major defect, Defect Code No. 08001.-The moisture shall be determined in accordance with Standard MIL-STD-650, Method 101.4 with methanol as a special solvent.

4.3.6 Determination of foreign matter.-The explosive composite shall be visually inspected for the presence of visible impurities and extraneous foreign matter.

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5. Preparation for Delivery

5.1 Preservation and packaging

5.1.1 Level A.-Unless otherwise specified by the procuring activity, the HMX shall be thoroughly mixed to form a slurry or cake containing not less than 10 percent by weight of a solution made as follows: 40 percent by weight of isopropyl alcohol and 60 percent water. The HMX shall be packed in this wet condition in bags made of double filled, gray cotton duck weighing not less than 12 ounces large enough to hold a maximum of 50 pounds (dry weight) of HMX mixture. The bags containing the wet HMX shall be securely closed with non-metallic cord and then placed in a rubber bag, rubberized cloth bag, or two bags, placed one inside the other and individually closed, made from 0.004 inch thick plyethylene film or bags made of suitable water-tight material. Not more than six of the smaller bags shall be placed in the larger bag. Each small bag shall contain HMX from one homogeneous lot.

5.2 Packing

5.2.1 Level A.-The large bag containing a maximum of 300 pounds dry weight of HMX shall be placed in a keg or drum complying with Interstate Commerce Commission Specification 5 or 5B or 21B as applicable, as required in the Code of Federal Regulations, Title 49 - Transportation, Parts 0-190.

5.3 Marking.-Containers shall be marked to insure safe handling as required by Code of Federal Regulations, Title 49 - Transportation, Parts 0-190. In addition, shipments shall be marked in accordance with Standard MIL-STD-129. Each container shall be clearly labeled with the lot number and net weight of its contents. When a container holds HMX composition from more than one lot, all lot numbers shall be shown on the label and each of the small bags inside the container shall be clearly and permanently labeled with the lot numbers of its contents.

6. Notes

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Description sheets shall be prepared for each lot in accordance with information specified in Standard MIL-STD-1171.

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6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification.

6.3 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.4 Intended use.-The HMX mixtures conforming to this specification are intended for use in 30mm ammunition.

6.5 Inspection equipment

6.5.1 Commercial inspection equipment is defined in AMC Regulation 702-2.

6.5.2 Contractor-designed equipment.-In the event that a contractor elects to design his own inspection equipment, details of the design (drawings, description, materials, etc.) shall be submitted to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3, Dover, N. J., for approval prior to fabrication and use. Approval of such designs may be delegated to the contract administration office for minor defects only.

CUSTODIAN:
ARMY - MU

PREPARING ACTIVITY
ARMY - MU

PROJECT NUMBER: 1376 - A062

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DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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COMMANDING OFFICER
United States Army
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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